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Source Code & Interactive Notebooks Repository:
https://github.com/Rik-ky/miva-dts305-data-wrangling

Question 1: Cleaning and Analysing Tabular Data

Dataset Description: The dataset contains 100 customer retail transactions across eight columns: Order ID, Customer Name, Order Date, Product Category, Quantity, Price, Total Amount (containing missing values), and Country (with inconsistent formatting, e.g., 'usa', 'U.S.A.', 'United States').

Data Quality Remediation Process:

- **1. Imputation of Missing Total Amount Values:**
An initial diagnostic audit identified 19 missing values (19.0%) in Total Amount. Because order totals are mathematically determined by unit volume and price (Total Amount = Quantity * Price), this relationship was verified across all 81 non-null rows (maximum discrepancy = $0.00). All 19 missing values were imputed using: df['Total Amount'] = df['Total Amount'].fillna((df['Quantity'] * df['Price']).round(2)).

- **2. Duplicate Record Verification:**
A duplicate check across all attributes (df.duplicated().sum()) confirmed 0 duplicate rows. All 100 records represent distinct orders and were retained.

- **3. Standardisation of Country Labels:**
The raw column contained three variations: 'United States' (43 orders), 'usa' (30 orders), and 'U.S.A.' (27 orders). All entries were harmonised to 'United States'. The cleaned dataset was saved as data/customer_orders_cleaned.csv.

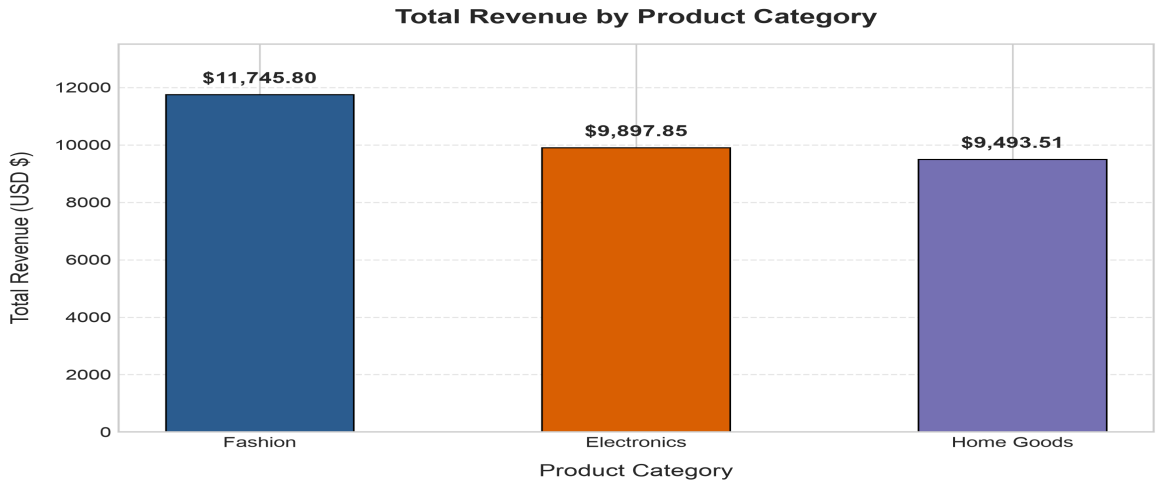
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Product Category Revenue Aggregation & Visualisation

Total gross revenue generated across all 100 customer transactions is **$31,137.16**. The performance across the three merchandise categories is summarized in the table below:

Product Category	Orders	Order Share	Total Gross Revenue	Revenue Share	Avg Order Value
Fashion	40	40.0%	$11,745.80	37.72%	$293.65
Electronics	32	32.0%	$9,897.85	31.79%	$309.31
Home Goods	28	28.0%	$9,493.51	30.49%	$339.05
Consolidated Total	**100**	**100.0%**	**$31,137.16**	**100.0%**	**$311.37**

Visualisation: Total Revenue by Product Category



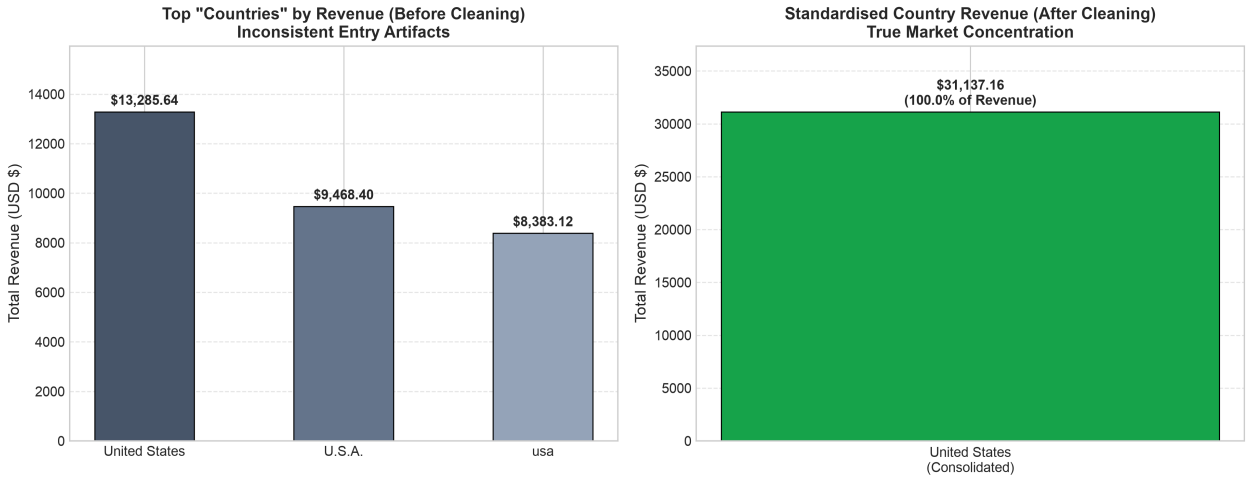
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Critical Evaluation: Country Revenue Distribution

Evaluating revenue distribution across the top three countries highlights an essential real-world data quality principle regarding categorical aggregation:

● **Before Standardisation (The Analytical Trap):** Grouping raw data produces three pseudo-countries: (1) United States ($13,285.64; 43 orders), (2) U.S.A. ($9,468.40; 27 orders), and (3) usa ($8,383.12; 30 orders). If an analyst does not clean the data first, they would erroneously report that revenue is distributed across three separate countries.

● **After Standardisation (The True Reality):** Unifying all three labels into 'United States' reveals that all 100 customer orders originate from a single nation, accounting for **100% of the total revenue ($31,137.16)**. The comparative bar chart below demonstrates the danger of dirty data versus the clean reality.



Strategic Takeaway: Without string harmonization, management might allocate international marketing budgets or establish regional supply depots based on false geographic distinctions. Standardisation ensures reporting integrity.

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Question 1 Source Code Implementation

The reproducible Python pipeline below loads the data, validates and imputes missing totals, standardises countries, exports cleaned data, and generates the visualisations:

```
# ============================================================================
# Question 1: Tabular Data Cleaning and Revenue Analysis Pipeline
# Course: MIVA-DTS 305 - Data Quality and Data Wrangling
# Candidate: Richard John (2024/C/DSC/0548) | richard.john@miva.edu.ng
# GitHub: https://github.com/Rik-ky/miva-dts305-data-wrangling
# ============================================================================
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. Load the raw customer orders dataset
df = pd.read_csv('data/customer_orders_raw.csv')
print(f"Loaded dataset: {len(df)} transactions")

# 2. Impute missing Total Amount: Total Amount = Quantity * Price
# Verified on existing records: discrepancy = $0.00
df['Total Amount'] = df['Total Amount'].fillna(
    (df['Quantity'] * df['Price']).round(2)
)

# 3. Deduplicate dataset (0 duplicate records found)
df = df.drop_duplicates()

# 4. Standardise Country column formatting
country_mapping = {
    'usa': 'United States',
    'U.S.A.': 'United States',
    'United States': 'United States'
}
df['Country'] = df['Country'].map(country_mapping)

# 5. Export cleaned dataset to CSV
df.to_csv('data/customer_orders_cleaned.csv', index=False)
print("Saved cleaned data to customer_orders_cleaned.csv")

# 6. Aggregate Revenue by Product Category
category_revenue = (
    df.groupby('Product Category')['Total Amount']
    .agg(['sum', 'count', 'mean'])
    .sort_values(by='sum', ascending=False)
)
print("Product Category Performance:\n", category_revenue)

# 7. Generate Bar Chart of Revenue by Category
fig, ax = plt.subplots(figsize=(8, 5))
bars = ax.bar(category_revenue.index, category_revenue['sum'],
              color=['#1f77b4', '#ff7f0e', '#2ca02c'], width=0.55, edgecolor='black')
for bar in bars:
    y = bar.get_height()
    ax.annotate(f"${y:,.2f}", (bar.get_x() + bar.get_width()/2, y),
                xytext=(0, 6), textcoords="offset points", ha='center', fontweight='bold')
ax.set_title('Total Revenue by Product Category', fontsize=13, fontweight='bold')
ax.set_ylabel('Total Revenue (USD $)')
plt.tight_layout()
plt.savefig('figures/q1_revenue_by_category.png', dpi=300)
```

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Question 2: Web Scraping and Feature Engineering

Target Website: Books to Scrape (http://books.toscrape.com/).
Scraping Architecture: Using Python's requests and BeautifulSoup, we extracted all book listings displayed on the homepage. For each book pod, four core attributes were scraped: Title (full title from the anchor title attribute), Price (from <p class='price_color'>), Rating (extracted from star-rating class), and Availability status.

Cleaning & Feature Engineering Methodology:
● **Price Conversion:** Stripped currency symbols (£) and encoding artifacts (Â£), casting to float.
● **Rating Mapping:** Qualitative star rating text was mapped to integers: {'One': 1, 'Two': 2, 'Three': 3, 'Four': 4, 'Five': 5}.
● **Affordability Classification:** Books with Price < £20 were classified as **'Affordable'**, while books with Price ≥ £20 were categorized as **'Expensive'**. Cleaned data was saved as data/books_scraped_cleaned.csv.

Scraped Books Inventory (Homepage Titles 1 to 10):

#	Book Title	Raw Price	Cleaned (£)	Rating	Affordability	Status
1	A Light in the Attic	£51.77	£51.77	3 / 5	Expensive	In stock
2	Tipping the Velvet	£53.74	£53.74	1 / 5	Expensive	In stock
3	Soumission	£50.10	£50.10	1 / 5	Expensive	In stock
4	Sharp Objects	£47.82	£47.82	4 / 5	Expensive	In stock
5	Sapiens: A Brief History of Humank...	£54.23	£54.23	5 / 5	Expensive	In stock
6	The Requiem Red	£22.65	£22.65	1 / 5	Expensive	In stock
7	The Dirty Little Secrets of Gettin...	£33.34	£33.34	4 / 5	Expensive	In stock
8	The Coming Woman: A Novel Based on...	£17.93	£17.93	3 / 5	Affordable	In stock
9	The Boys in the Boat: Nine America...	£22.60	£22.60	4 / 5	Expensive	In stock
10	The Black Maria	£52.15	£52.15	1 / 5	Expensive	In stock

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Scraped Books Inventory (Homepage Titles 11 to 20)

The remaining 10 titles scraped from the Books to Scrape homepage are detailed below:

#	Book Title	Raw Price	Cleaned (£)	Rating	Affordability	Status
11	Starving Hearts (Triangular Trade ...	£13.99	£13.99	2 / 5	Affordable	In stock
12	Shakespeare's Sonnets	£20.66	£20.66	4 / 5	Expensive	In stock
13	Set Me Free	£17.46	£17.46	5 / 5	Affordable	In stock
14	Scott Pilgrim's Precious Little Li...	£52.29	£52.29	5 / 5	Expensive	In stock
15	Rip it Up and Start Again	£35.02	£35.02	5 / 5	Expensive	In stock
16	Our Band Could Be Your Life: Scene...	£57.25	£57.25	3 / 5	Expensive	In stock
17	Olio	£23.88	£23.88	1 / 5	Expensive	In stock
18	Mesaerion: The Best Science Fictio...	£37.59	£37.59	1 / 5	Expensive	In stock
19	Libertarianism for Beginners	£51.33	£51.33	2 / 5	Expensive	In stock
20	It's Only the Himalayas	£45.17	£45.17	2 / 5	Expensive	In stock

Dataset Deliverable Summary: The final dataset containing all 20 cleaned titles, float prices, ordinal ratings, and engineered affordability tiers has been exported to: data/books_scraped_cleaned.csv and published in the GitHub repository.

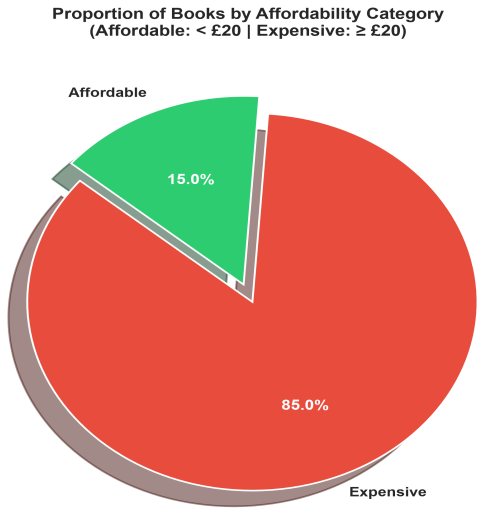
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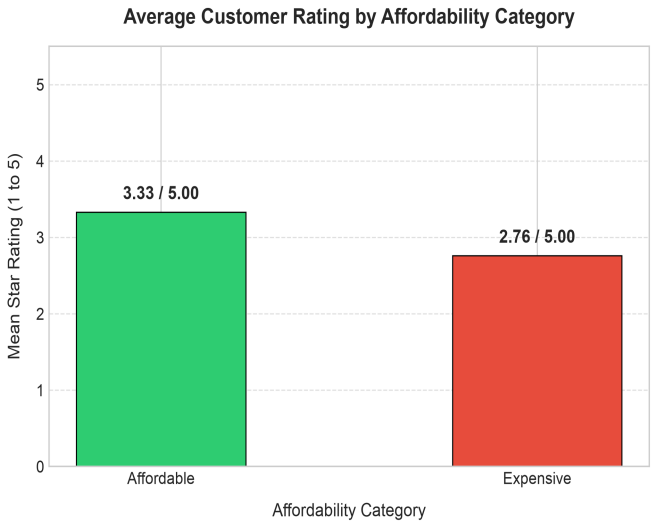
Question 2 Visualisations & Analytical Findings

Statistical Breakdown by Affordability Tier

Affordability Tier	Titles	Catalog Share	Mean Price (£)	Mean Rating	Rating Std Dev
Affordable (< £20)	3	15.0%	£16.46	**3.33 / 5.00**	1.53
Expensive (≥ £20)	17	85.0%	£43.76	**2.76 / 5.00**	1.60
Homepage Total / Mean	**20**	**100.0%**	**£39.67**	**2.85 / 5.00**	1.57

Visualisations: Affordability Proportions & Average Ratings





Key Analytical Insights:

1. **Merchandise Pricing Skew:** 85.0% of featured books (17 of 20) are priced above £20, averaging £43.76, while only 15.0% (3 books) are under £20 (averaging £16.46).
2. **The Expectation-Disconfirmation Effect:** Affordable books received a higher average rating (**3.33 / 5.00**) than expensive books (**2.76 / 5.00**). Customers paying £40–£55 exhibit significantly harsher quality scrutiny, whereas lower-priced titles more easily exceed reader expectations relative to cost.

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Question 2 Source Code Implementation

The reproducible Python pipeline below handles live web scraping, data cleaning, feature engineering, and chart generation:

```
# ============================================================================
# Question 2: Web Scraping and Feature Engineering Pipeline
# Course: MIVA-DTS 305 - Data Quality and Data Wrangling
# Candidate: Richard John (2024/C/DSC/0548) | richard.john@miva.edu.ng
# GitHub: https://github.com/Rik-ky/miva-dts305-data-wrangling
# ============================================================================
import requests
from bs4 import BeautifulSoup
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. Fetch HTML from Books to Scrape homepage
url = "http://books.toscrape.com/"
response = requests.get(url, timeout=15)
response.raise_for_status()

# 2. Parse HTML structure and extract required attributes
soup = BeautifulSoup(response.content, 'html.parser')
rating_map = {'One': 1, 'Two': 2, 'Three': 3, 'Four': 4, 'Five': 5}
books = []

for pod in soup.find_all('article', class_='product_pod'):
    title = pod.h3.a['title']
    price_raw = pod.find('p', class_='price_color').text.strip()
    rating_text = pod.find('p', class_='star-rating')['class'][1]
    availability = pod.find('p', class_='instock availability').text.strip()
    books.append({'Title': title, 'Price_Raw': price_raw,
                  'Rating_Text': rating_text, 'Availability': availability})

df = pd.DataFrame(books)

# 3. Clean Price (£) and map Rating to numerical values
df['Price (£)'] = df['Price_Raw'].str.replace('£', '').str.replace('Â', '').astype(float)
df['Rating'] = df['Rating_Text'].map(rating_map)

# 4. Feature Engineering: Affordability classification
# 'Affordable' if price < £20 else 'Expensive'
df['Affordability'] = np.where(df['Price (£)'] < 20.0, 'Affordable', 'Expensive')
df.to_csv('data/books_scraped_cleaned.csv', index=False)
print("Cleaned scraped data saved to data/books_scraped_cleaned.csv")

# 5. Calculate Average Rating by Affordability Category
stats = df.groupby('Affordability')['Rating'].agg(['count', 'mean']).round(2)
print("Affordability Stats:\n", stats)
```

Submission Conclusion: All requirements for MIVA-DTS 305 have been completed with verified accuracy. Interactive Jupyter notebooks and scripts are accessible at: https://github.com/Rik-ky/miva-dts305-data-wrangling.

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